# Physics II

## Set no. 1

E - exam type question / task

simple harmonic oscillations

$$\vec{F} = -k\vec{r}$$

$$F_x = -kx$$

$$ma_x = -kx$$

$$m\frac{d^2x}{dt^2} = -kx$$

$$\boxed{\frac{d^2x}{dt^2} + \frac{k}{m}x = 0}$$

$$x(t) = A\sin(\omega t + \varphi) \; ; \; \omega = \sqrt{\frac{k}{m}}$$

linear (1D) wave equation

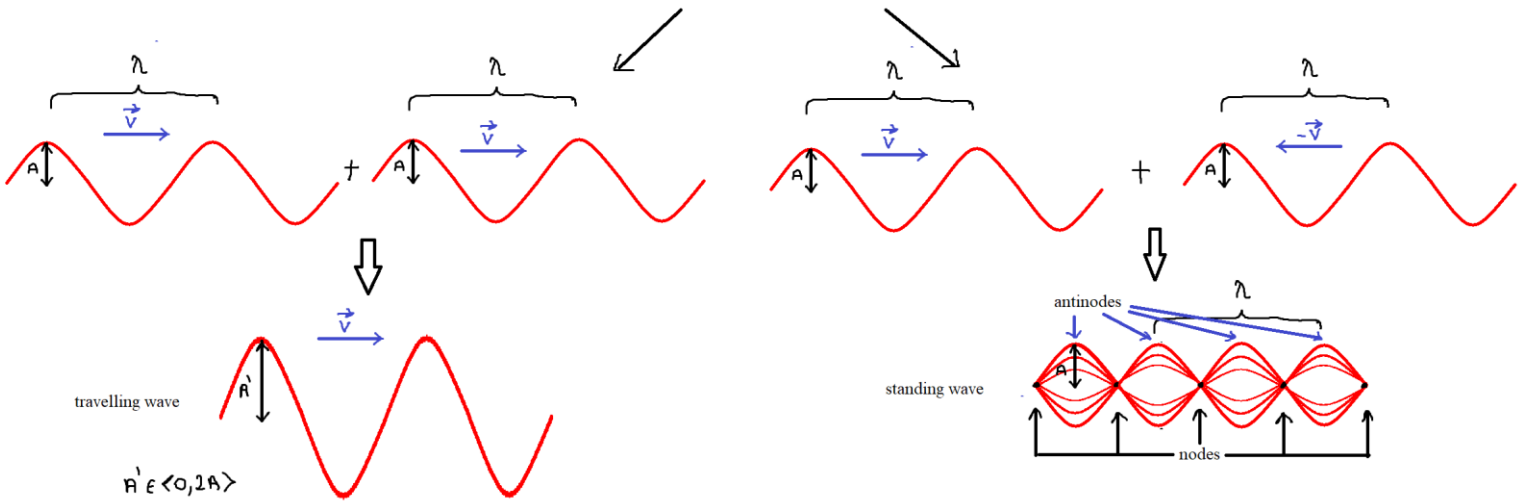
$$\boxed{\frac{d^2y}{dx^2} = \frac{1}{v^2}\frac{d^2y}{dt^2}}$$

$$y = y(x \pm vt)$$

e.g

$$y(x,t) = A\sin(kx - \omega t)$$

sinusoidal wave superposition (interference)



1. A particle moves along the $x$ axis from $x = -A$ to $x = A$, in simple harmonic motion with period $T$. At time $t = 0$ it is at $x = A$. Please:

   a) write the particle position x as a function of time, x(t)
   b) determine the position and velocity $v_x$ when $t = 0.75T$.

2. A particle with mass $m = 0.02$ kg moves in simple harmonic motion according to $x(t) = 2\cos(50t)$, where $x$ is in meters and $t$ is in seconds. Please calculate:
   a) maximum velocity, $v_{max}$
   b) spring constant, $k$
   c) maximum potential energy.

3. A $m = 0.25$-kg block oscillates on the end of the spring with a spring constant of $k = 200$ N/m. Please find the amplitude of the oscillation if the oscillation is started by elongating the spring $x = 0.15$ m and giving the block a speed of $v = 3$ m/s.

4. The period of a simple pendulum is $T_1 = 1$ s on Earth. Please determine the period $T_2$ of this pendulum when brought to a planet where gravitational acceleration is one-tenth that on Earth.

5. A particle moves along the $x$-axis from $x = -A$ to $x = A$ in damped harmonic motion, with spring constant k = 80 N/m and damping coefficient b = 0.2 kg/s. Please:
   a) write the equation of motion and its solution, i.e. x(t), assuming that x(t = 0) = A,
   b) calculate the frequency of the particle's motion,
   c) determine the time after which the particle loses 25% of its initial energy.

6. Please check that a function $y(x,t) = A\sin(kx - \omega t)$ satisfy the linear wave equation. Determine wave speed as a function of a wave number k and an angular frequency ω.

7. A wave moving along the x axis is described by:

$$y(x,t) = 5 \cdot e^{-(x+4t)^2}$$

   where x is in meters and t is in seconds. Please:
   a) prove that the given wave function satisfy the wave equation,
   b) determine the direction of the wave motion and the speed of the wave.

E 8. When a particular wire is vibrating with a frequency of $f = 4$ Hz, a transverse wave of wavelength $\lambda = 60$ cm is produced. Please determine the speed of wave pulses along the wire.

9. A sinusoidal wave traveling in the positive x direction has an amplitude of $A = 15$ cm, a wavelength of $\lambda = 40$ cm, and a frequency of $f = 8$ Hz. The vertical displacement of the

medium at t = 0 and x = 0 is also y = 15 cm. Please find the angular wave number *k*, period *T*, angular frequency $\omega$, and speed *v* of the wave and then write the wave function.

10. Two waves in one string are described by the wave functions:

$$y_1 = 3.0 \cos(4.0x - 1.6t)$$

$$y_2 = 4.0 \sin(5.0x - 2.0t)$$

where y and x are in centimeters and t is in seconds. Please find the superposition of the waves:

a) at x = 1 cm, t = 1 s,

b) at x = 1 cm, t = 0.5 s,

c) for all points and for any time (the superposition wave function of these two waves).

E 11. Two sinusoidal waves of the same frequency and the same amplitude *A* propagate in the same direction in the same medium. As a result of their interference, the amplitude of the wave at a certain point is *A*' = 1.93*A*. Please find the phase difference between these two waves.

E 12. Standing waves are produced by the interference of two travelling sinusoidal waves, each of frequency *f* = 200 Hz. The distance from the 2nd node to the 5th node is *d* = 60 cm. Please calculate:

a) the wavelength,
b) the speed of each of the two original waves.

13. A transverse wave on a string is described by the expression:

$$y = (0.120 \text{ m}) \sin(\pi x/8 + 4\pi t)$$

a) Please determine the transverse speed and acceleration of the string at t = 0.2 s for the point on the string located at x = 1.6 m.

E b) What are the wavelength, period, and speed of propagation of this wave?

E 14. Transverse waves with a speed of v = 50 m/s are to be produced in a taut string. A L = 5.00-m length of string with a total mass of *m* = 0.6 kg is used. What is the required tension?

E 15. A piano string having a mass per unit length $\mu = 5 \cdot 10^{-3}$ kg/m is under a tension of $F_T$ = 1350 N. Please find the speed with which a wave travels on this string